

Sophora (Japanese Pagoda Tree)

Styphnolobium japonicum

Species background: Native to China and Korea, historically planted near temples and palaces; sometimes reclassified in scientific literature under the name *Styphnolobium japonicum*. It's one of the few street trees that flowers in August, when almost nothing else nearby is blooming -- a useful clue for identifying it from a distance.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Sophora (Japanese Pagoda Tree): Count sophora's smooth leaflets and find its green twigs.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, crayons.

Teacher background

Count the smooth little leaflets on one leaf. Check if a young twig still looks green -- that means it's still making food through its bark.

Procedure

- 1 Engage: Ask students to count something small and repeating nearby.
- 2 Explore: Count the leaflets on a real or pictured compound leaf together.
- 3 Explain: Explain that green twigs mean the tree is still making food through its bark, not just its leaves.
- 4 Art: Count the smooth little leaflets, then draw a row of matching ovals.
- 5 Evaluate: Student can count leaflets correctly and point out a green twig.

Discussion questions

What color would you expect a twig to be, and why might green be a surprise?

Assessment

Correct leaflet count and identification of a green twig.

Extension

Look for green twigs on other young trees during your walk.

Sophora (Japanese Pagoda Tree): Explain why sophora's late-summer bloom timing is unusual.

Standards connection: NGSS 3-LS4-3; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

This tree blooms in late summer, when almost no other tree is flowering -- that means the bees have it all to themselves.

Procedure

- 1 Engage: Ask what month students think most trees bloom, then reveal this one blooms later.
- 2 Explore: If in bloom, observe the creamy flower clusters; otherwise look at photos.
- 3 Explain: Discuss why blooming later, when there's less competition for pollinators, could be an advantage.
- 4 Art: This tree blooms late, when almost nothing else is flowering -- write a few lines about being different from the crowd, like this tree's late blooms.
- 5 Evaluate: Student explains in one sentence why late blooming might help this tree get noticed by bees.

Discussion questions

Can you think of a time being different from everyone else around you turned out to be an advantage?

Assessment

Correct one-sentence explanation of the late-bloom advantage.

Extension

Note which other trees on your walk are blooming (or not) at the same time.

GRADE BAND 6-8

Sophora (Japanese Pagoda Tree): Explain niche flowering timing as a pollinator-competition strategy, and correct the tree's origin misconception.

Standards connection: NGSS MS-LS2-2; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, sketching materials.

Teacher background

Blooming in late summer, when fewer competitors are still flowering, means more pollinator attention for this species. Despite its common name, it's native to China, not Japan -- it was introduced to Europe through Japanese trade ports.

Procedure

- 1 Engage: Why might a common name for a species sometimes be misleading about where it actually comes from?
- 2 Explore: Research how this tree's 'Japanese pagoda tree' name came about despite its Chinese origin.
- 3 Explain: Connect its niche late-summer bloom timing to reduced competition for pollinators.
- 4 Art: Create a still-life sketch of a sophora flower cluster in late summer, paired with one sentence about a time you did something on a different timeline than everyone else.
- 5 Evaluate: Write a paragraph correcting the origin misconception and explaining the niche-timing advantage.

Discussion questions

What other plants or products have common names that don't match their true origin?

Assessment

Paragraph correctly addresses both the origin misconception and the bloom-timing strategy.

Extension

Research one other plant with a misleading common name.

Sophora (Japanese Pagoda Tree): Evaluate bloom timing as a competitive strategy independent of native range.

Standards connection: NGSS HS-LS2-6; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, research sources, field-notebook materials.

Teacher background

Blooming in late summer reduces competition for pollinators. Research this species' origin as an 18th-century botanical introduction from China, and evaluate how bloom timing can function as a competitive strategy independent of a plant's native range.

Procedure

- 1 Engage: Does a plant's evolved bloom timing 'know' it's competing with other species, or is this simply the outcome of selection pressure over time?
- 2 Explore: Research this species' 18th-century introduction to Europe as a botanical curiosity from Chinese trade routes.
- 3 Explain: Evaluate niche flowering phenology as a general competitive strategy, using this species as a case study.
- 4 Art: Research botanical illustration plates from 18th-century plant-hunting expeditions, then create your own field-notebook-style page with sketch, notes, and location data.
- 5 Evaluate: Submit the field-notebook page and a short written evaluation of bloom-timing as a competitive strategy.

Discussion questions

How might climate change affecting bloom timing across many species change the competitive landscape this tree currently benefits from?

Assessment

Rubric covering historical research accuracy and depth of the competitive-strategy evaluation.

Extension

Compare sophora's niche-timing strategy to littleleaf linden's role as a late-nectar source for bees.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.